

Supplement Sx

Molecular Solar Thermal (MOST) as Photon-Gated Thermal Storage

A Mode-Path Matrix Perspective on Photochemically Addressable Heat

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

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Learning Objectives

- LO-Sx.1** Explain how MOST systems function as photon-gated thermal storage and map the charge–store–discharge cycle onto the Mode-Path Matrix framework as a radiation→metastable→conduction mode conversion.
- LO-Sx.2** Compute the penetration depth $\delta = 1/\alpha$ for a given MOST solution and use the regime indicator δ/d to classify a microchannel as optically thin, transitional, or optically thick.
- LO-Sx.3** Implement the 1.5D depth-sliced flow model to predict volumetric state-of-charge $\bar{X}_{\text{out}}$ and dead-zone indicator X_{min} , and demonstrate why high absorption efficiency does *not* imply uniform volumetric charging.
- LO-Sx.4** Define and compute the MOST module figures of merit— E_{m} , E_{sys} , k_{photo} , $\bar{X}_{\text{out}}$, X_{min} , and $P_{\text{th,del}}$ —and compare them against PCM, hot-water, and thermochemical storage benchmarks.
- LO-Sx.5** Design a spectral-splitting hybrid architecture that routes UV to a MOST charging channel and visible/NIR to a PV cell, using dichroic optics (Ch. 8) and the MPM to manage all thermal paths.
- LO-Sx.6** Execute the microfluidic MOST design exercise: from irradiance map through depth-sliced kinetics to outlet conversion, using LightTools for ray tracing and Python for photo-kinetics.

Abbreviations and Nomenclature

Abbreviation	Meaning
BL	Beer–Lambert (absorption law)
CPD	Cyclobutane Pyrimidine Dimer (DNA photodamage product)
DR	Design Rule
FOM	Figure of Merit
LED	Light-Emitting Diode
MOST	Molecular Solar Thermal (energy storage)
MPM	Mode-Path Matrix
NBD/QC	Norbornadiene/Quadricyclane (photoswitch pair)

Abbreviation	Meaning
PCM	Phase-Change Material
PV	Photovoltaic
SoC	State of Charge
TPD	Thermal Photonics & Diffusionics
TPV	Thermophotovoltaic
WE	Worked Example

Symbol	Meaning	SI Unit
α	Absorption coefficient $(\ln 10) \varepsilon C$	cm^{-1}
C	Molar concentration	mol L^{-1}
d	Channel depth (optical thickness)	cm
δ	Optical penetration depth $1/\alpha$	cm
E_m	Active-material enthalpy density	MJ kg^{-1}
E_{sys}	System-level stored energy density	MJ kg^{-1}
$\varepsilon(\lambda)$	Molar absorptivity	$\text{L mol}^{-1} \text{cm}^{-1}$
η_Q	Effective photoisomerization quantum yield	—
η_{HX}	Heat exchanger effectiveness	—
Γ	Mode dominance ratio $G_{\text{rad}}/G_{\text{cond}}$	—
I_0	Incident irradiance at channel surface	W cm^{-2}
k_{photo}	Photo-rate constant	s^{-1}
L	Illuminated channel length	cm
N_A	Avogadro's number (6.022×10^{23})	mol^{-1}
$P_{\text{th,del}}$	Delivered thermal power	W
Q	Volumetric flow rate	$\text{cm}^3 \text{s}^{-1}$
τ	Metastable storage lifetime	s
t_{res}	Residence time in illuminated zone	s
U_{vol}	Volumetric SoC nonuniformity $\sigma_X/\bar{X}$	—
w	Channel width	cm
X	State of charge (fraction photo-converted)	—
$\bar{X}_{\text{out}}$	Volume-average outlet SoC	—
X_{min}	Minimum-layer outlet SoC (dead-zone indicator)	—

Numbering convention: Design rules use DR-Sx.n. Worked examples use WE-Sx.n. Learning objectives use LO-Sx.n.

Supplement Roadmap and Deliverables

A simulation-ready optical–microfluidic design methodology for MOST thermal storage modules, framed within the Mode-Path Matrix, with a 1.5D depth-sliced model that correctly captures volumetric charging stratification.

The supplement proceeds as follows: §Sx.1 frames the thermal engineering gap that MOST addresses. §Sx.2 maps the MOST cycle onto the MPM framework. §Sx.3 develops the penetration depth criterion. §Sx.4 presents the 1.5D depth-sliced simulation pipeline. §Sx.5 collects figures of merit. §Sx.6 works a quantitative example exposing the optically thick paradox. §Sx.7 describes spectral-splitting hybrid architectures. §Sx.8 gives a design exercise for optical staff. §Sx.9 links MOST to specific book chapters. Figure Sx.1 shows the workflow spine.

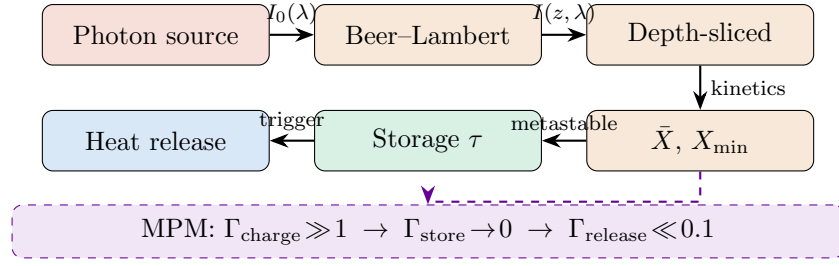


Figure Sx.1: Supplement Sx roadmap. The pipeline runs from photon source through Beer–Lambert absorption and depth-sliced kinetics to thermal release. The dashed box shows the MPM mode classification at each stage.

Sx.1 Why This Supplement: The Thermal Storage Gap

Pain Point

Thermal photonics excels at *controlling* heat flows—spectral selectivity (Ch. 8), near-field enhancement (Ch. 3), radiative cooling (Ch. 9), selective emitters (Ch. 10), and thermophotovoltaics (Ch. 10)—but practical thermal systems also require **storage** that is (i) long-duration, (ii) compact, and (iii) dispatchable on demand without continuous insulation losses. Phase-change materials leak heat. Batteries store electricity, not heat. What if the storage medium itself were *optically addressable*?

Molecular Solar Thermal (MOST) energy storage systems answer this question. A photoreponsive molecule absorbs a photon, undergoes a conformational change to a strained high-energy isomer, and remains locked in that state for months to years. An external trigger (heat pulse, acid, or catalyst) releases the stored enthalpy as thermal energy [1, 3].

The most recent advance is the Dewar pyrimidone system of Nguyen *et al.* [1], which achieves an active-material enthalpy density $E_m > 1.6$ MJ/kg—roughly double a lithium-ion battery on a thermal basis—with storage lifetimes exceeding three years and water solubility enabling flow-based architectures.

Engineering re-interpretation for optics staff

In MOST, optics plays the role of *front-end power electronics*: it shapes where and which photons are deposited to maximize (a) chemical conversion, (b) spatial uniformity, and (c) safety/robustness, while minimizing parasitic bulk heating. The optical simulation is **not complete** until it produces $\bar{X}_{\text{out}}$ and X_{min} maps—not just irradiance maps.

Sx.2 MOST in the Mode-Path Matrix Framework

The Mode-Path Matrix (Ch. 4) classifies every thermal path by the mode dominance ratio $\Gamma = G_{\text{rad}}/G_{\text{cond}}$ and routes each path to the appropriate design tool: $\Gamma < 0.1 \rightarrow$ Part II, $0.1 < \Gamma < 10 \rightarrow$ Ch. 9, $\Gamma > 10 \rightarrow$ Part III.

MOST introduces a *temporal dimension* to this classification: the system traverses three distinct Γ regimes in sequence during one operating cycle. Figure Sx.2 shows this as a Mode-Path Matrix network diagram.

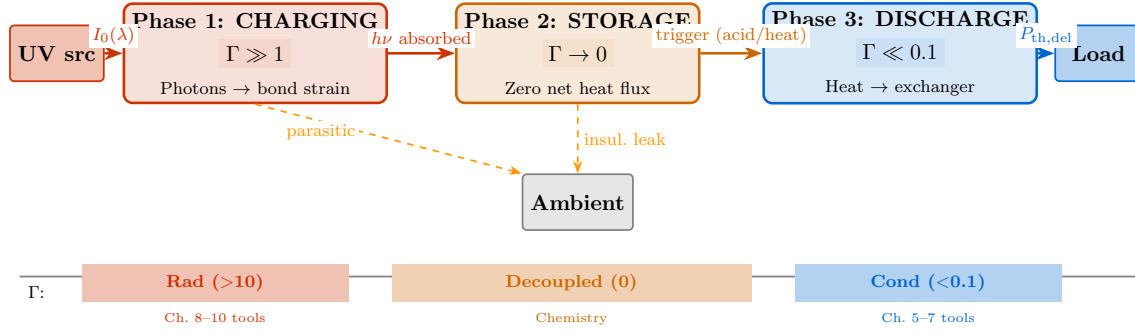


Figure Sx.2: Mode-Path Matrix network diagram for a MOST operating cycle. The system traverses three Γ regimes: radiation-dominated charging ($\Gamma \gg 1$, red), fully decoupled storage ($\Gamma \rightarrow 0$, amber), and conduction-dominated discharge ($\Gamma \ll 0.1$, blue). Dashed orange arrows indicate parasitic loss paths to ambient. The bottom bar shows the Γ classification with corresponding TPD chapter routing.

The three phases are:

1. **Charging** ($\Gamma_{\text{charge}} \gg 1$): Photons are the sole energy input. The optical path controls SoC. Thermal conduction carries only parasitic waste heat.
2. **Storage** ($\Gamma_{\text{store}} \rightarrow 0$): No net heat flux in either mode. Energy is locked in molecular bond strain—a perfect thermal insulator on the stored-energy degree of freedom.
3. **Discharge** ($\Gamma_{\text{release}} \ll 0.1$): An exothermic reversion releases heat locally, conducted into a heat exchanger. Radiation is negligible at discharge temperatures ($\Delta T \sim 50\text{--}100$ K).

Table Sx.3 shows the formal MPM entry.

Table Sx.3: Mode-Path Matrix entry for a MOST module. Each phase occupies a separate row because Γ changes with operational state.

Phase	Path	G_{cond} (W/K)	G_{rad} (W/K)	G_{conv} (W/K)	Γ	Tool
Charge	UV src $\rightarrow$ fluid	parasitic	G_{phot}	—	$\gg 1$	This Supp. + Ch. 8
Store	Fluid (tank)	~ 0	~ 0	—	N/A	Chemistry
Release	Fluid $\rightarrow$ HX $\rightarrow$ load	G_{HX}	negl.	$G_{\text{conv,HX}}$	$\ll 0.1$	Ch. 5–7

The “photonic thermal conductance” during charging is defined as:

$$G_{\text{phot}} = \frac{\dot{m} E_{\text{sys}} \bar{X}_{\text{out}}}{\Delta T_{\text{source-fluid}}}, \quad (\text{Sx.1})$$

allowing MOST to be slotted into the MPM alongside conventional thermal elements.

Trilogy Connection: Photon-Gated Thermal Switch

The MOST storage phase ($\Gamma \rightarrow 0$) is a *molecular-scale thermal diode* with infinite rectification ratio (cf. Ch. 6 GST-based diodes with $\mathcal{R} \sim 3\text{--}50$). Energy flows in *neither* direction along *any* mode during storage—fundamentally different from phase-change diodes, which modulate G_{cond} but cannot halt it entirely.

Sx.3 Physical Regimes: The Penetration Depth Criterion

Absorption follows Beer–Lambert:

$$I(z, \lambda) = I_0(\lambda) \exp[-\alpha(\lambda) z], \quad \alpha(\lambda) = \ln(10) \varepsilon(\lambda) C. \quad (\text{Sx.2})$$

The optical penetration depth and the dimensionless regime indicator are:

$$\delta \equiv \frac{1}{\alpha}, \quad \boxed{\text{Regime indicator} = \frac{\delta}{d}}. \quad (\text{Sx.3})$$

Figure Sx.3 illustrates the three regimes visually.

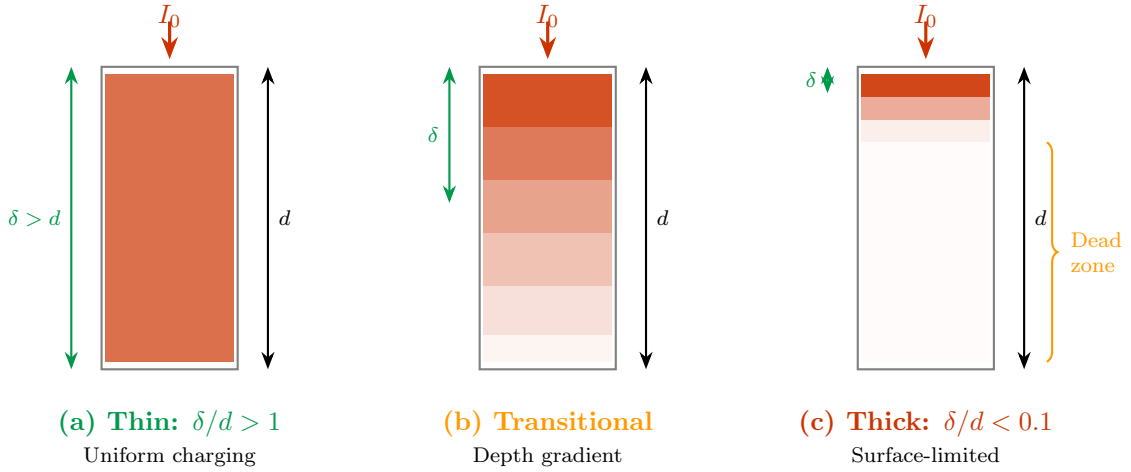


Figure Sx.3: Three optical regimes in a MOST microchannel (cross-sectional view, illumination from top). Color intensity represents local photon flux. (a) Optically thin: uniform penetration. (b) Transitional: significant gradient. (c) Optically thick: photons absorbed within $\sim 3\delta$ of the surface; bulk of the channel is a dead zone.

DR-Sx.1 — Penetration Depth Regime Classification

Classify the microchannel charging regime by δ/d :

- $\delta/d > 1$: **Optically thin**. 1D model is adequate.
- $0.1 < \delta/d < 1$: **Transitional**. Use the 1.5D depth-sliced model.
- $\delta/d < 0.1$: **Optically thick**. The 1D model is unsafe. Must use depth-sliced model or 2D CFD.

The Optically Thick Paradox

When $\alpha d \gg 1$, the channel absorbs $\sim 100\%$ of incident photons— giving the misleading impression of efficient charging. However, **absorption efficiency** $\neq$ **volumetric charging efficiency**. All photon energy is deposited within $\sim \delta$ of the surface; deeper layers remain uncharged. WE-Sx.1 (§Sx.6) shows that a channel absorbing $>99.97\%$ of photons can yield $\bar{X}_{\text{out}} \approx 0.19$ —a $4.2\times$ overestimate by the naive 1D model.

Sx.4 Simulation Pipeline: From Irradiance to Outlet SoC

Sx.4.1 1D Baseline Model (Optically Thin Limit)

For $\delta/d > 1$, the absorbed photon flux over depth d is:

$$\Phi_{\text{abs}} = \frac{I_0}{hc/\lambda} [1 - \exp(-\alpha d)]. \quad (\text{Sx.4})$$

The photo-rate constant (column-averaged) is:

$$k_{\text{photo}} = \eta_Q \frac{\Phi_{\text{abs}} \times 1000}{C d N_A} \quad (\text{s}^{-1}), \quad (\text{Sx.5})$$

where the factor 1000 converts L to cm^3 . The SoC evolves as:

$$X(t) = 1 - \exp(-k_{\text{photo}} t). \quad (\text{Sx.6})$$

For flow: $t = t_{\text{res}} = w d L / Q$ with Q the volumetric flow rate.

Steady-State Limitation

The 1D model distributes absorbed photons uniformly across the entire column. Valid *only* when $\delta/d > 1$. For $\delta/d < 1$, it assigns photons to molecules that never receive them.

Sx.4.2 1.5D Depth-Sliced Flow Model

Divide the channel into N slices ($N \geq 20$), each of thickness $\Delta = d/N$. For slice i at depth z_i :

$$I_{\text{in},i} = I_0 \exp(-\alpha i \Delta), \quad (\text{Sx.7})$$

$$\Phi_i = \frac{I_{\text{in},i}}{hc/\lambda} [1 - \exp(-\alpha \Delta)], \quad (\text{Sx.8})$$

$$k_i = \eta_Q \frac{\Phi_i}{N_\Delta}, \quad N_\Delta = \frac{C \Delta N_A}{1000}, \quad (\text{Sx.9})$$

$$X_i = 1 - \exp(-k_i t_{\text{res}}). \quad (\text{Sx.10})$$

Volume statistics:

$$\bar{X}_{\text{out}} = \frac{1}{N} \sum_i X_i, \quad X_{\text{min}} = \min_i X_i, \quad U_{\text{vol}} = \frac{\sigma_X}{\bar{X}_{\text{out}}}. \quad (\text{Sx.11})$$

DR-Sx.2 — Minimum Safe SoC Specification

Specify *both*: $\bar{X}_{\text{out}} \geq 0.80$ and $X_{\text{min}} \geq 0.50$. If X_{min} fails while $\bar{X}$ passes, depth stratification is severe: reduce C , reduce d , or add dual-side illumination.

Sx.4.3 Thermal Release and Heat Exchanger

Ideal thermal power: $P_{\text{th,ideal}} = \dot{m} E_{\text{sys}} \bar{X}_{\text{out}}$. Delivered power: $P_{\text{th,del}} = \eta_{\text{HX}} P_{\text{th,ideal}}$.

The system energy density includes dilution:

$$E_{\text{sys}} = \frac{C M_w}{1000 \rho_{\text{sol}}} E_{\text{m}}. \quad (\text{Sx.12})$$

Sx.4.4 In-Line Optical SoC Metrology

A dual-wavelength absorbance ratio sensor closes the control loop: $A(\lambda) = \varepsilon(\lambda) \ell C$, calibrated vs. X .

Sx.5 Figures of Merit

Table Sx.4: MOST module figures of merit for Thermal Photonics comparison.

FOM	Symbol	Meaning	Typical	Pyrimidone
Material enthalpy	E_m (MJ/kg)	Pure compound ceiling	0.1–0.5 ^a	>1.6
System energy	E_{sys} (MJ/kg)	Incl. solvent dilution	10^{-2} – 10^{-1}	$\sim 0.18^b$
Storage lifetime	τ (yr)	No insulation losses	0.01–1	>3
Depth ratio	δ/d	Stratification indicator	design-dep.	WE-Sx.1
Volume SoC	$\bar{X}_{\text{out}}$	Usable stored fraction	0.5–0.95	tgt ≥ 0.8
Min SoC	X_{min}	Dead-zone detector	>0.5	tgt ≥ 0.5
Photo-rate	k_{photo} (s ⁻¹)	Charging speed	10^{-3} – 10^{-1}	case-dep.
Thermal power	$P_{\text{th,del}}$ (W)	Usable heat rate	10^0 – 10^3	appl.-dep.

^aPrior MOST: NBD/QC ~ 0.4 , azobenzenes ~ 0.2 MJ/kg. ^bAt $C = 0.1$ mol/L, $M_w \approx 110$ g/mol.

Table Sx.5 benchmarks MOST against conventional storage.

Table Sx.5: MOST vs. conventional thermal storage benchmarks.

Technology	E_{sys} (MJ/kg)	τ	Dispatch?	Key limitation
Hot water ($\Delta T = 60$ K)	0.25	hours	Yes	Heat leak
PCM (paraffin)	0.15–0.25	hours	Yes	Supercooling
Thermochem. (CaO)	1.0–2.0	months	Yes	Sintering, cost
MOST (pyrimidone)	$\sim 0.18^c$	years	Yes	UV-only, scale

^cAt demonstrated concentrations. Never compare E_m (>1.6) against Li-ion electrical energy without Carnot correction (DR-Sx.5).

Sx.6 Worked Example: The Optically Thick Paradox (WE-Sx.1)

WE-Sx.1: Depth Stratification in a Pyrimidone Microchannel

Purpose. Demonstrate quantitatively that a channel absorbing >99.97% of photons can yield $\bar{X}_{\text{out}} \approx 0.19$ at the residence time the 1D model predicts $X = 0.80$.

Parameters (replace with measured values before design):

Table Sx.6: WE-Sx.1 input parameters.

Parameter	Value	Note
λ_0	300 nm	Pyrimidone peak
I_0	50 mW/cm ²	UV-LED module
$\varepsilon(\lambda_0)$	5×10^3 L mol ⁻¹ cm ⁻¹	Conservative
C	0.10 mol/L	Moderate
d	200 μ m = 0.020 cm	Std. microfluidic
w	1 mm = 0.10 cm	—
η_Q	0.50	Placeholder

Sx.6.1 Step A: Regime Classification

$$\alpha = 2.303 \times 5000 \times 0.10 = 1151.5 \text{ cm}^{-1}, \quad (\text{Sx.13})$$

$$\delta = 1/\alpha = 8.68 \text{ } \mu\text{m}, \quad (\text{Sx.14})$$

$$\delta/d = 8.68/200 = 0.043 < 0.1 \implies \textbf{Optically thick.} \quad (\text{Sx.15})$$

Channel absorptance: $\mathcal{A} = 1 - \exp(-23.03) > 0.9999$.

Sx.6.2 Step B: 1D Model (baseline for comparison)

Photon energy at 300 nm: $E_{\text{ph}} = hc/\lambda = 6.626 \times 10^{-19}$ J. Incident photon flux: $\Phi_0 = I_0/E_{\text{ph}} = 7.55 \times 10^{16}$ photons s⁻¹ cm⁻².

Molecules per cm² column: $\mathcal{N}_{\text{tot}} = C d N_A/1000 = 1.204 \times 10^{18}$ cm⁻².

$$k_{\text{photo}}^{\text{1D}} = 0.50 \times \frac{7.55 \times 10^{16}}{1.204 \times 10^{18}} = 0.0314 \text{ s}^{-1}. \quad (\text{Sx.16})$$

Time to $X = 0.80$:

$$t_{\text{req}}^{\text{1D}} = -\ln(0.20)/0.0314 = 51.2 \text{ s}. \quad (\text{Sx.17})$$

Sx.6.3 Step C: 1.5D Depth-Sliced Model

$N = 10$ slices, $\Delta = 20 \text{ } \mu\text{m}$, $\alpha\Delta = 2.303$, in-slice absorptance $f_{\text{abs}} = 0.90$. Each slice transmits only 10% of its input—a factor-of-10 attenuation per slice.

Table Sx.7: WE-Sx.1: Depth-sliced results at $t_{\text{res}} = 51.2$ s.

Slice i	z_i (μ m)	$I_{\text{in},i}/I_0$	k_i (s ⁻¹)	X_i
0	10	1.000	0.282	1.000
1	30	0.100	0.0282	0.763
2	50	0.010	0.00282	0.134
3	70	0.001	2.82×10^{-4}	0.014
4	90	10^{-4}	2.82×10^{-5}	0.001
5–9	110–190	$< 10^{-5}$	~ 0	≈ 0
Volume-average:				$\bar{X}_{\text{out}} = \mathbf{0.191}$
Minimum SoC:				$X_{\text{min}} \approx \mathbf{0}$

Figure Sx.4 visualizes the dramatic depth stratification.

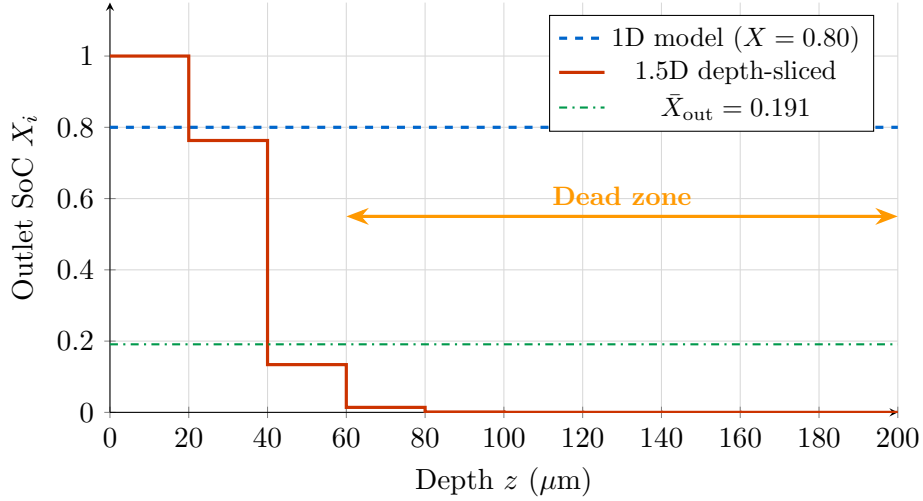


Figure Sx.4: WE-Sx.1: Outlet SoC versus depth for the 1D model (blue dashed, uniform $X = 0.80$) and the 1.5D depth-sliced model (red steps). The volume-average (green dash-dot, $\bar{X} = 0.191$) is $4.2\times$ lower than the 1D prediction. The dead zone (orange bracket) encompasses $\sim 70\%$ of the channel volume.

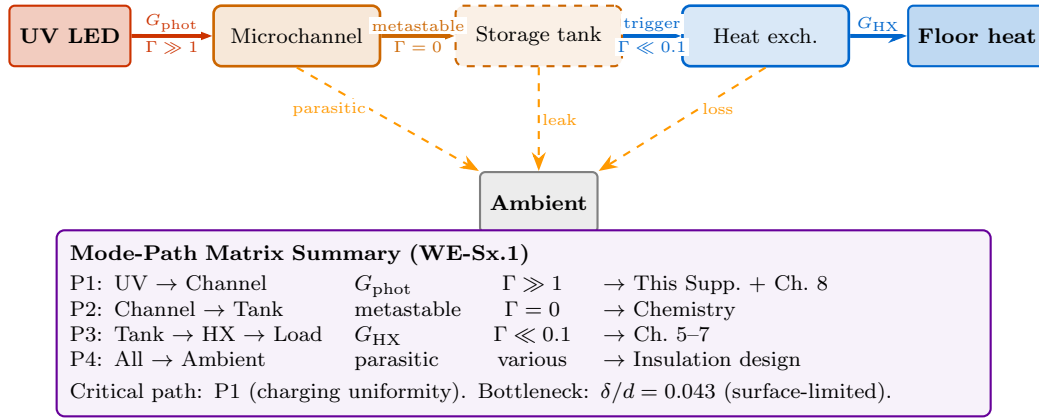


Figure Sx.5: Mode-Path Matrix network diagram for WE-Sx.1. Solid arrows show the three-phase energy path (charge → store → discharge → load); line color and Γ label classify each edge. Dashed orange arrows indicate parasitic loss paths. The summary box shows the formal MPM with chapter routing.

Sx.6.4 Step D: Design Remediation

Four optics lever arms to fix the stratification:

- Reduce concentration** to increase δ . Target: $\delta/d \geq 0.2$, requiring $C \leq 1/(2.303 \times 5000 \times 5 \times 0.020) \approx 0.0087$ mol/L. Trade-off: lower E_{sys} .
- Reduce channel depth d** . At $C = 0.10$ mol/L, need $d \leq 5\delta \approx 43$ μm . Reduces per-channel throughput.
- Dual-side illumination**. Halves effective optical depth, giving $2\delta/d$. Requires transparent substrate.
- Increase irradiance**. Speeds charging but does *not* fix the depth profile—the exponential attenuation ratio is unchanged.

DR-Sx.3 — Optics Lever Arms for MOST Charging

Report three optics parameters: (1) I_0 (irradiance level), (2) δ/d (penetration-to-depth ratio), (3) $U_{\text{spatial}} = \sigma_I/\bar{I}$ (lateral uniformity). Reporting only irradiance is insufficient.

Sx.7 Spectral-Splitting Hybrid Architecture

The pyrimidone absorbs at 300 nm—only $\sim 5\%$ of the AM1.5 solar spectrum. A spectral-splitting architecture (Ch. 8, §08.3) routes each band to its optimal converter. Figure Sx.6 shows the optical layout; Figure Sx.7 shows the corresponding MPM network.

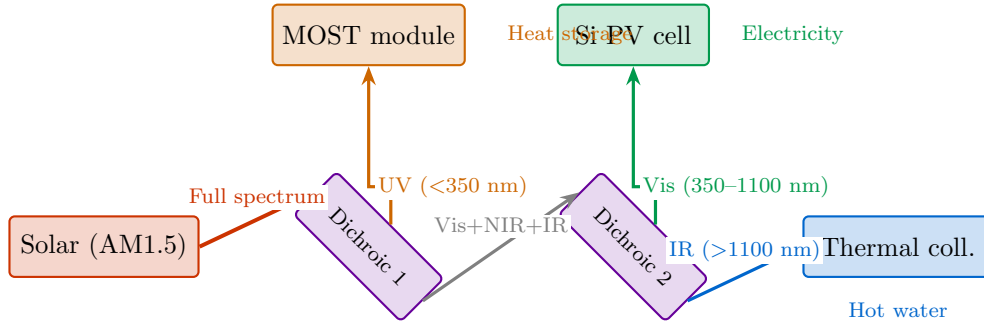


Figure Sx.6: Spectral-splitting MOST+PV+Thermal hybrid: two dichroic mirrors route three spectral bands to their optimal converters. UV (<350 nm) charges the MOST fluid; visible (350–1100 nm) generates electricity; NIR/IR (>1100 nm) heats water directly.

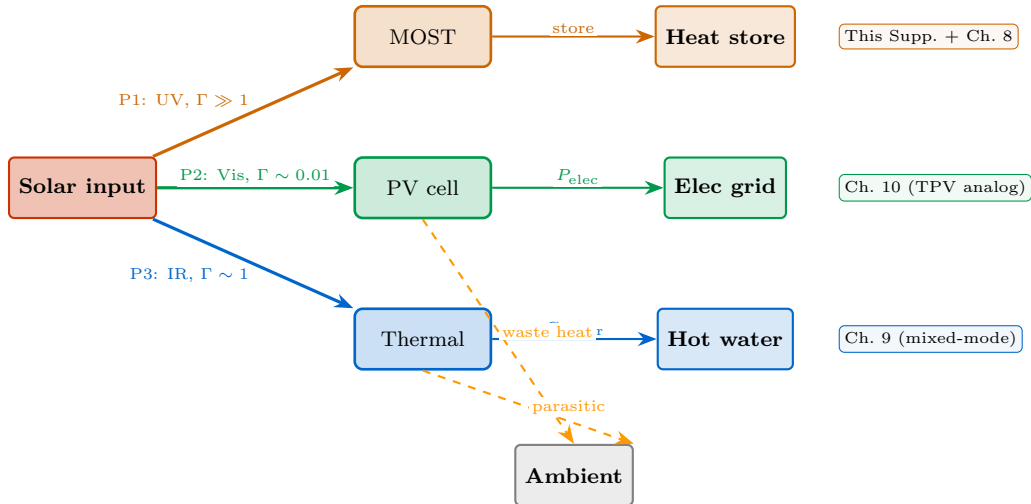


Figure Sx.7: Mode-Path Matrix network for the spectral-splitting hybrid. Three parallel paths (P1–P3) from a single solar node, each classified by Γ and routed to the appropriate TPD chapter. Dashed arrows: parasitic losses.

Trilogy Connection: Spectral Routing = Mode Routing

In Book 1, spectral filters route wavelengths to different focal planes. In Book 2, dichroics separate signal from background. Here in Book 3, dichroics route spectral bands to different *thermal modes*: UV $\rightarrow$ chemical storage, visible $\rightarrow$ electricity, IR $\rightarrow$ thermal. Spectral routing *is* mode routing in the MPM.

Sx.8 Design Exercise

Design Exercise Sx (4–8 hours, optics-facing)

Objective: Design a microfluidic MOST charging module achieving $\bar{X}_{\text{out}} \geq 0.80$, $X_{\text{min}} \geq 0.50$ at $Q = 0.5$ mL/min with $U_{\text{spatial}} \leq 0.05$.

Tasks:

1. LightTools: model UV-LED + non-imaging optics; export $I(x, y)$.
2. Check δ/d (DR-Sx.1). If < 0.1 , redesign *before* proceeding.
3. Python: implement 1.5D depth-sliced model ($N \geq 20$).
4. Report: $X(z)$ cross-section, $\bar{X}$ and X_{min} vs. Q , energy budget.
5. Propose two redesigns improving X_{min} by $\geq 2\times$.

Listing 1: Depth-sliced MOST charging model.

```

1  import numpy as np
2
3  # --- Input parameters ---
4  lambda0 = 300e-7      # cm (300 nm)
5  I0 = 0.050           # W/cm^2
6  eps_mol = 5000.0     # L/(mol cm)
7  C = 0.10             # mol/L
8  d = 0.020            # cm (200 um)
9  eta_Q = 0.50
10 w = 0.10             # cm
11 Q = 8.33e-3          # cm^3/s (0.5 mL/min)
12 N_slices = 20
13
14 h_planck = 6.626e-34; c_light = 3e10; N_A = 6.022e23
15 E_ph = h_planck * c_light / lambda0 # J/photon
16
17 # --- Derived quantities ---
18 alpha = np.log(10) * eps_mol * C     # cm^-1
19 delta_pen = 1.0 / alpha              # cm
20 Delta = d / N_slices                # slice thickness
21 print(f"delta/d = {delta_pen/d:.4f}")
22
23 # --- Depth-sliced model ---
24 N_Delta = C * Delta * N_A / 1000.0   # molecules/cm^2 per slice
25 f_abs = 1.0 - np.exp(-alpha * Delta)
26
27 k = np.zeros(N_slices)
28 for i in range(N_slices):
29     I_in = I0 * np.exp(-alpha * i * Delta)
30     Phi_i = (I_in / E_ph) * f_abs
31     k[i] = eta_Q * Phi_i / N_Delta
32
33 # --- Sweep illuminated length ---
34 L_arr = np.linspace(1, 500, 200)
35 results = []
36 for L_ill in L_arr:
37     t_res = w * d * L_ill / Q
38     X = 1.0 - np.exp(-k * t_res)
39     results.append((L_ill, t_res, np.mean(X), np.min(X)))

```

Sx.9 Connection to TPD Chapters

Table Sx.8: MOST module design tasks mapped to TPD book chapters.

Design Task	Chapter	Tool
Mode classification (Γ)	Ch. 4	MPM
Near-field enhancement ($d < 100$ nm)	Ch. 3	$G_{\text{NF}}(d)$
Dichroic spectral splitting	Ch. 8	Cold-mirror (§08.3)
Radiative cooling of PV cell	Ch. 9	Atm. window emitter
Parasitic heat management	Ch. 9	Alg. 09.A
TPV back-reflector analogy	Ch. 10	Photon recycling
Microfluidic HX design	Ch. 5–7	Conduction tools
Thermal diode analogy	Ch. 6	Switchable G_{cond}
Failure modes	Ch. 12	I/M/F/T framework
Adaptive SoC feedback	Ch. 15	Sensor-driven control

Bio-Inspired Photochemistry → Engineered Thermal Storage

The pyrimidone [2+2] cycloaddition is the *same photochemistry* that causes UV-induced DNA damage (CPDs). Evolution treats this as pathology; MOST engineers treat it as energy storage. This inversion suggests systematically mining the photobiology literature for new MOST candidates: any photoproduct that is (a) metastable, (b) exothermic upon reversion, and (c) synthetically accessible is a potential thermal storage material.

Design Rules Summary

Table Sx.9: Complete set of MOST supplement design rules.

Rule	Section	Description
DR-Sx.1	§Sx.3	Classify by δ/d : > 1 thin, 0.1–1 transitional, < 0.1 thick. Use 1.5D model for $\delta/d < 1$.
DR-Sx.2	§Sx.4.2	Specify both $\bar{X}_{\text{out}} \geq 0.80$ and $X_{\text{min}} \geq 0.50$.
DR-Sx.3	§Sx.6	Report: I_0 , δ/d , and U_{spatial} .
DR-Sx.4	§Sx.7	For solar MOST, use spectral splitting (UV→MOST, Vis→PV). Single-band wastes $>95\%$ of spectrum.
DR-Sx.5	§Sx.5	Distinguish E_{m} (pure solute) from E_{sys} (system incl. dilution). Never compare E_{m} against Li-ion without Carnot correction.

Looking Forward

MOST is at TRL 2–3, but the 2026 pyrimidone breakthrough is a step-change. Near-term priorities: (i) red-shifting absorption toward 400–500 nm; (ii) demonstrating >1000 cycles; (iii) kilogram-scale synthesis; (iv) prototype spectral-splitting hybrid module. The optical design tools of Chapters 3, 8, and 9 are directly applicable to all four.

Problems

1. **Regime classification.** A MOST fluid has $\varepsilon(365 \text{ nm}) = 1.2 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1}$ at $C = 0.05 \text{ mol/L}$, $d = 100 \text{ }\mu\text{m}$. (a) Compute α , δ , δ/d . (b) Classify the regime. (c) How

- does δ/d change at $C = 0.20$ mol/L? (d) What d gives $\delta/d = 0.5$ at $C = 0.05$?
2. **1D vs. 1.5D comparison.** Using Problem 1 values with $I_0 = 30$ mW/cm², $\eta_Q = 0.6$: (a) Compute $k_{\text{photo}}^{\text{1D}}$ and t_{req} for $X = 0.80$. (b) 1.5D model ($N = 20$): compute $\bar{X}$ at t_{req} . (c) Overestimate factor? (d) At what δ/d does the overestimate drop below 10%?
 3. **System energy density.** Pyrimidone derivative ($M_w = 124$, $E_m = 1.45$ MJ/kg) at $C = 0.20$ mol/L. (a) Compute E_{sys} . (b) Compare vs. hot water ($\Delta T = 60$ K). (c) At what C does E_{sys} match the hot-water value?
 4. **Spectral budget.** Estimate AM1.5G power in the 280–320 nm band. (a) Integrated irradiance. (b) UV power on 1 m² collector. (c) Compare vs. 365 nm UV-LED at 50 mW/cm² over 1 m².
 5. **MPM for building heating.** Rooftop collector (1 m²), 50 L tank ($C = 0.15$ mol/L), in-floor HX ($\eta_{\text{HX}} = 0.70$). (a) Construct MPM with ≥ 4 paths. (b) Compute Γ for each. (c) Estimate winter-evening $P_{\text{th,del}}$ at 2 L/min discharge. (d) Compare vs. 50 L hot water at $\Delta T = 50$ K.
 6. **Dual-side illumination.** (a) Derive $I(z) = (I_0/2)[\exp(-\alpha z) + \exp(-\alpha(d-z))]$. (b) Show minimum at $z = d/2$ improves from $\exp(-\alpha d)$ to $\text{sech}(\alpha d/2)$. (c) For WE-Sx.1 parameters, compute X_{min} improvement.

References

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Supplement Sx of “Thermal Photonics: A Unified Framework for Engineering Heat Flux”

Book 3 of the Open-Source Engineering Optics Trilogy

Cross-references: Book 1 Ch. 6 (thin-film dichroics); Book 2 Ch. 8 (spectral discrimination); TPD Ch. 4 (MPM), Ch. 6 (thermal diodes), Ch. 8 (spectral engineering), Ch. 9 (mixed-mode), Ch. 10 (TPV)